

Module 1

Linux & Shell Basics

Learning objectives

- Open a terminal and identify your working directory
- Navigate with `cd`, `pwd`, and `ls`
- Create, copy, move, and delete files safely
- View text with `cat`, `less`, `head`, and `tail`
- Use `man` and `--help` to explore commands

Prerequisites

- Unix-like environment (Linux, macOS, or WSL2)
- Ability to open a terminal and type commands
- No prior shell experience required

Module learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Terminal & shell] --> B[Paths & navigation]

B --> C[Files & directories]

C --> D[View & edit text]

D --> E[man and --help]

E --> F[Exercises & Module 2]

Topics build from terminal basics toward project navigation

Why this matters for verification

- Run simulators, scripts, and makefiles from the shell
- Move between src/, tb/, and build/ directories
- Inspect logs, RTL, and test output as text files

Terminal & navigation

Shell vs terminal

- Terminal emulator — the window you type in
- Shell (bash, zsh) — interprets commands
- Prompt shows user, host, and current directory

Module 1 self-check

```
$ ./scripts/module1.sh --check
```

Terminal: module1_check

```

[0;36m=====
[0;36mModule 1: Environment self-check
[0;36m=====

[0;32m[OK] [0m pwd: /home/yongfu/proj/universal-verification-methodology/learn_unix_git
[0;32m[OK] [0m ls available
[0;32m[OK] [0m cat available
[0;32m[OK] [0m less available
[0;32m[OK] [0m man available
[0;32m[OK] [0m module1 directory exists: /home/yongfu/proj/universal-verification-methodology/
[0;32m[OK] [0m sample_for_viewing.txt exists
[0;32m[OK] [0m sample_repo (navigation) exists

[0;32m[OK] [0m All required checks passed.
```


Navigation commands

```
pwd  
ls  
ls -la  
cd ~/projects  
cd ..  
cd -
```

RTL-style project layout

```
$ cd module1/examples/navigation/sample_repo && pwd && ls -la && cd  
src && pwd && ls
```

Terminal: navigation_sample_repo

```
/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module1/examples/navigation/  
total 0  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 .  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 ..  
-rwxrwxrwx 1 yongfu yongfu 210 May 28 12:03 README.txt  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 build  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 src  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 tb  
/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module1/examples/navigation/
```

Files & viewing

Files and directories

- `mkdir -p` for nested directories
- `cp` and `mv` for copy and rename
- `rm` and `rm -r` — use carefully; avoid `rm -rf` mistakes

Exercise scaffold

```
mkdir ~/unix_practice  
mkdir ~/unix_practice/notes ~/unix_practice/projects  
cd ~/unix_practice && pwd
```

Viewing files

```
$ head -n 5 module1/examples/viewing/sample_for_viewing.txt && tail -n  
3 module1/examples/viewing/sample_for_viewing.txt
```

Terminal: viewing_head_tail

Line 01: This is a sample file for Module 1 viewing practice.
Line 02: Use head, tail, and less to navigate.
Line 03: head -n N shows the first N lines.
Line 04: tail -n N shows the last N lines.
Line 05: less lets you scroll and search.
Line 28: Practice in a safe directory first.
Line 29: This file has enough lines for head/tail/less practice.
Line 30: End of sample_for_viewing.txt.

Create practice dirs (script)

```
./scripts/module1.sh --scaffold
```

Help & practice

Getting help

- `man ls` — full manual (Space, q to quit)
- `ls --help` — short option summary
- Tab completion for commands and paths
- History with `↑` and `↓` (Ctrl+R in bash)

Repo resources

- module1/examples/ — hands-on labs
- module1/CHECKLIST.md — exercises
- ./scripts/module1.sh — check & demo

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Exercises (your turn)

- Hello terminal — `pwd`, `ls`, `ls -la`
- Directory navigation — `unix_practice/`
- File operations — `cp`, `mv`, `rm` with `nano`
- Viewing — `head`, `tail`, `less` on a long file
- Help — `man ls`, tab completion

Summary & next steps

- You can navigate and manage files from the shell
- Run `./scripts/module1.sh --check` before Module 2
- Next — Module 2: Filesystem, Permissions & Environment